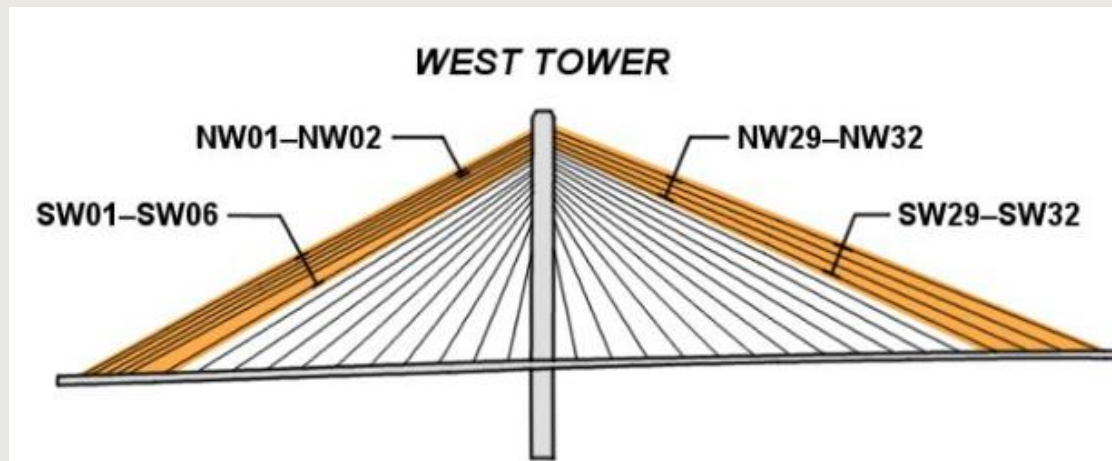


Vibration and Damping Analysis of a Bridge Stay Cable

C263F PROJECT MIDTERM PRESENTATION

Completed Tasks



$$\mathbf{f}_k^{\text{highRe}} = -\frac{1}{2}\rho C_D A_k \|\mathbf{u}_k^{\text{rel}}\| \mathbf{u}_k^{\text{rel}}.$$

01. SET UP A 2D CABLE MODEL

Give an inclined straight initial position.
Implemented a 2D edge-by-edge Cosserat cable model, computing forces segment-wise.

02. APPLY FORCE

Gravity is applied to every node.
Introduce initial tension.
Update high-Re drag function to 2D (wind).
Add lift force.
Add a viscous force.

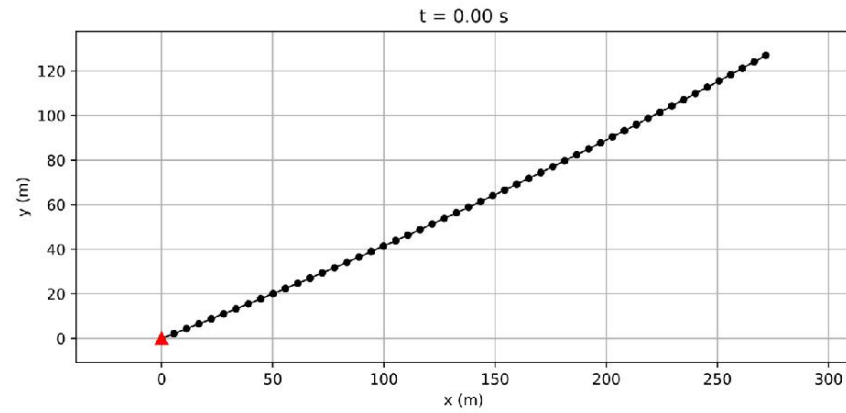
03. TIME ITERATION

Forces = $F_s + F_b + F_g + F_h + F_v + F_l$
J_Forces = $J_s + J_b + J_h + J_v + J_l$

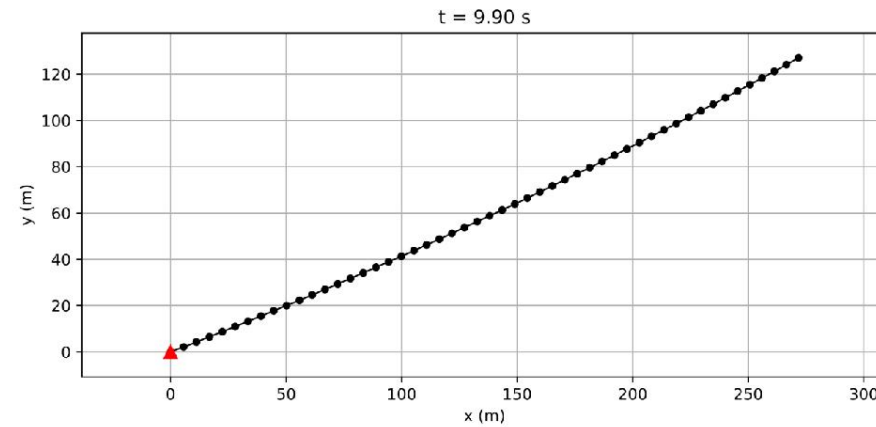
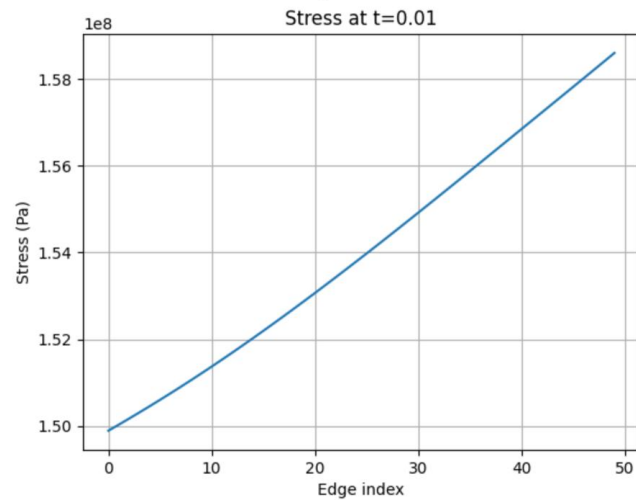
04. STRESS ANALYZE

Estimate cable stress & "potential broken point"

Preliminary Results



Newton iter 50, residual = 3.251e-04
Newton hit max_iter=50, residual=3.251e-04
Max stress = 158597134.14479637 Pa at edge 49



Current time: 2.6099999999999883

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.792e+03 N
Fv (damping): 0.000e+00 N
Fl (lift): 8.377e+02 N

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.791e+03 N
Fv (damping): 4.208e+02 N
Fl (lift): 8.372e+02 N

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.791e+03 N
Fv (damping): 4.208e+02 N
Fl (lift): 8.372e+02 N

Current time: 3.239999999999975

Fs (stretch): 2.573e+06 N
Fb (bend): 2.991e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.792e+03 N
Fv (damping): 0.000e+00 N
Fl (lift): 8.377e+02 N

Fs (stretch): 2.573e+06 N
Fb (bend): 2.990e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.788e+03 N
Fv (damping): 4.630e+02 N
Fl (lift): 8.365e+02 N

Fs (stretch): 2.573e+06 N
Fb (bend): 2.990e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.788e+03 N
Fv (damping): 4.630e+02 N
Fl (lift): 8.365e+02 N

Further Goals

STEP 1

Deck vibration
(time-dependent boundary motion)

STEP 2

Time-history tracking

Multiple dampers
(tower-side + deck-side)

Frequency-domain evaluation